

Menglin Zhu

Curriculum Vitae with References

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Research Interest

- Probing quantum/energy materials atom by atom using advanced (S)TEM:
 - Multislice electron ptychography ▪ *Operando* microscopy ▪ 4D-STEM ▪ Spectroscopy
- Modeling atomistic behavior and phase transitions using DFT and *ab initio* MD simulations.
- Extracting insights from microscopy data using statistical analysis and machine learning

Education and Training

- 2023–present **Postdoctoral Researcher**, *Massachusetts Institute of Technology*, Prof. James M. LeBeau.
2018–2023 **Ph.D. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2014–2017 **B.S. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2013–2014 **Materials Science and Engineering**, *Nanjing University of Aeronautics and Astronautics*.

Honors & Awards

- 2026 **Albert Crewe Award** of Microscopy Society of America to recognize distinguished contributions to the field of microscopy in the physical sciences of an early career scientist
- 2025 **PARADIM user proposal #366** *Probing Dynamic Polar Structures in Relaxors*, 10 days of microscope time at Cornell PARADIM center
- 2023 **Postdoctoral Scholar** of Microscopy Society of America, awarded as a postdoctoral researcher for the paper submitted for the Microscopy and Microanalysis conference 2023
- 2022 **Student Scholar** of Microscopy Society of America, awarded as a Ph.D. candidate for the paper submitted for the Microscopy and Microanalysis conference 2022
- 2021 **Presidential Fellow** of Ohio State University, the most prestigious award given by the Graduate School to outstanding Ph.D. candidates
- 2017 **The Mars Fontana Scholarship** by Ohio State University, awarded to the most outstanding senior student in the Department of Material Science and Engineering
- 2016 **Summer Research Fellow** of Ohio State University
Markworth-Woolley Scholarship by Ohio State University

Publications

Citations: 1439, h-Index: 17, *Equal Contribution, #Correspondance

- [49] ²⁰²⁶ **Menglin Zhu***, Michael Xu*, Zishen Tian*, Colin Gilgenbach, Daniel Drury, Bridget R Denzer, Ching-Che Lin, Deokyoung Kang, Lane W Martin, and James M LeBeau. *Operando Electron Microscopy of Nanoscale Electronic Devices on Nonconductive Substrates. Microscopy and Microanalysis*, volume 32, page ozag007, DOI:10.1093/mam/ozag007.
- [48] ²⁰²⁶ **Menglin Zhu***, Michael Xu*, Yubo Qi, Colin Gilgenbach, Jieun Kim, Jiahao Zhang, Bridget R Denzer, Lane W Martin, Andrew M Rappe, and James M LeBeau. Bridging experiment and theory of relaxor ferroelectrics with multislice electron ptychography. *Science*, volume 392, pages 519–523, DOI:10.1126/science.ads6023.

- [47] ²⁰²⁶ Zishen Tian, **Menglin Zhu**, Jaegyu Kim, Piush Behera, Michael Xu, Thomas J Lee, Ching-Che Lin, Sreekeerthi Pamula, Archana Raja, Hao Pan, Jieun Kim, James M LeBeau, and Lane W Martin. Unleashing the electromechanical response of ferroelastic domain reorganization in mixed-phase tetragonal ferroelectric multilayers. *Adv. Mater.*, page e18417, DOI:10.1002/adma.202518417.
- [46] ²⁰²⁶ Colin Gilgenbach, **Menglin Zhu**, and James M LeBeau. Phaser: A unified and extensible framework for fast electron ptychography. *Npj Comput. Mater.*, pages 1–13, DOI:10.1038/s41524-026-01956-8.
- [45] ²⁰²⁵ **Menglin Zhu***, Michael Xu*, Yu Yun, Liyan Wu, Or Shafir, Colin Gilgenbach, Lane W Martin, Ilya Grinberg, Jonathan E Spanier, and James M LeBeau. Insights into chemical and structural order at planar defects in Pb₂MgWO₆ using multislice electron ptychography. *ACS Nano*, DOI:10.1021/acsnano.4c14833.
- [44] ²⁰²⁵ **Menglin Zhu***, Michael Xu*, Louis Alaerts, Hao Pan, Colin Gilgenbach, Geoffroy Hautier, Lane W Martin, and James M LeBeau. Direct determination of antiferroelectric-to-ferroelectric phase transition pathways in PbZrO₃ with Operando Electron Microscopy. *arXiv [cond-mat.mtrl-sci]*, DOI: arXiv2509.07194.
- [43] ²⁰²⁵ **Menglin Zhu***, Joseph Lanier*, Sevim Polat Genlik*, Jose G Flores, Victor da Cruz Pinha Barbosa, Mohit Randeria, Patrick M Woodward, Maryam Ghazisaeidi, Fengyuan Yang, and Jinwoo Hwang. Emergent ferromagnetism at LaFeO₃/SrTiO₃ interface arising from a strain-induced spin-state transition. *Adv. Mater. Interfaces*, volume 12, page 2500169, DOI:10.1002/admi.202500169.
- [42] ²⁰²⁵ Jith Sarker, Prachi Garg, Abrar Rauf, Ahsior Rahman Nirjhar, Hsien-Lien Huang, **Menglin Zhu**, A F M Anhar Uddin Bhuiyan, Lingyu Meng, Hongping Zhao, Jinwoo Hwang, Eric Osei-Agyemang, Saquib Ahmed, and Baishakhi Mazumder. Microscopic and spectroscopic investigation of (Al_xGa_{1-x})₂O₃ films: Unraveling the impact of growth orientation and aluminum content. *Adv. Mater. Interfaces*, volume 12, page 2301016, DOI:10.1002/admi.202301016.
- [41] ²⁰²⁵ Hao Pan, Liyan Wu, John Carroll, **Menglin Zhu**, Zishen Tian, Dongfang Chen, Hongrui Zhang, Xianzhe Chen, Xiaoxi Huang, Irina Baraban, Sreekeerthi Pamula, Cedric J G Meyers, R Ramesh, Kathleen Coleman, Brendan Hanrahan, James M LeBeau, Jonathan E Spanier, and Lane W Martin. Highly tunable relaxors developed from antiferroelectrics. *Adv. Mater.*, page e2505376, DOI:10.1002/adma.202505376.
- [40] ²⁰²⁵ Jieun Kim, Yubo Qi, Abinash Kumar, Yun-Long Tang, Michael Xu, Hiroyuki Takenaka, **Menglin Zhu**, Zishen Tian, Ramamoorthy Ramesh, James M LeBeau, Andrew M Rappe, and Lane W Martin. Size-driven phase evolution in ultrathin relaxor films. *Nat. Nanotechnol.*, pages 1–9, DOI:10.1038/s41565-025-01863-x.
- [39] ²⁰²⁵ I-Hsuan Kao, Junyu Tang, Gabriel Calderon Ortiz, **Menglin Zhu**, Sean Yuan, Rahul Rao, Jiahan Li, James H Edgar, Jiaqiang Yan, David G Mandrus, Kenji Watanabe, Takashi Taniguchi, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Unconventional unidirectional magnetoresistance in heterostructures of a topological semimetal and a ferromagnet. *Nat. Mater.*, pages 1–9, DOI:10.1038/s41563-025-02175-0.
- [38] ²⁰²⁵ Deokyoung Kang, Xue-Zeng Lu, Megha Acharya, Sajid Husain, Isaac Harris, Piush Behera, Ching-Che Lin, Ella Banyas, Alex Smith, Francesco Ricci, **Menglin Zhu**, Bridget R Denzer, Tanguy Terlier, Shu Wang, Tae Yeon Kim, Lucas Caretta, Douglas Natelson, James M LeBeau, Jeffrey B Neaton, Ramamoorthy Ramesh, James M Rondinelli, and Lane W Martin. Bulk-Rashba effect with suppressed spin relaxation in a polar phase of Bi_{1-x}In_{1+x}O₃. *Adv. Mater.*, page e2504684, DOI:10.1002/adma.202504684.
- [37] ²⁰²⁵ Aaditya Bhat, Colin Gilgenbach, Junghwa Kim, Michael Xu, **Menglin Zhu**, and James M LeBeau. Sensitivity of multislice electron ptychography to point defects: A case study in SiC. *Ultramicroscopy*, volume 280, page 114282, DOI:10.1016/j.ultramic.2025.114282.

- [36] ²⁰²⁴ **Menglin Zhu**, Joseph Lanier, Jose Flores, Victor da Cruz Pinha Barbosa, Daniel Russell, Becky Haight, Patrick M Woodward, Fengyuan Yang, and Jinwoo Hwang. Structural degeneracy and formation of crystallographic domains in epitaxial LaFeO₃ films revealed by machine-learning assisted 4D-STEM. *Sci. Rep.*, volume 14, page 4198, DOI:10.1038/s41598-024-54661-1.
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- [34] ²⁰²⁴ Jith Sarker, Prachi Garg, Abrar Rauf, Ahsiur Rahman Nirjhar, Hsien-Lien Huang, **Menglin Zhu**, A F M Anhar Uddin Bhuiyan, Lingyu Meng, Hongping Zhao, Jinwoo Hwang, Eric Osei-Agyemang, Saquib Ahmed, and Baishakhi Mazumder. Microscopic and spectroscopic investigation of (Al_xGa_{1-x})₂O₃ films: Unraveling the impact of growth orientation and aluminum content. *Adv. Mater. Interfaces*, DOI:10.1002/admi.202301016.
- [33] ²⁰²⁴ Hao Pan*, **Menglin Zhu***, Ella Banyas, Louis Alaerts, Megha Acharya, Hongrui Zhang, Jiyeob Kim, Xianzhe Chen, Xiaoxi Huang, Michael Xu, Isaac Harris, Zishen Tian, Francesco Ricci, Brendan Hanrahan, Jonathan E Spanier, Geoffroy Hautier, James M LeBeau, Jeffrey B Neaton, and Lane W Martin. Clamping enables enhanced electromechanical responses in antiferroelectric thin films. *Nat. Mater.*, DOI:10.1038/s41563-024-01907-y.
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- [28] ²⁰²³ Xiaolei Guo, Hsien-Lien Huang, **Menglin Zhu**, Karthikeyan Hariharan, Szu-Chia Chien, Ngan Huynh, Jinwoo Hwang, Wolfgang Windl, Christopher D Taylor, Eric J Schindelholz, and Gerald S Frankel. Interstitial elements created via metal 3D printing. *Mater. Today*, volume 66, pages 92–104, DOI:10.1016/j.mattod.2023.04.020.
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- [23] ²⁰²² Olivia G Licata, **Menglin Zhu**, Jinwoo Hwang, and Baishakhi Mazumder. Nanoscale chemistry and ion segregation in zirconia-based ceramic at grain boundaries by atom probe tomography. *Scr. Mater.*, volume 213, page 114603, DOI:10.1016/J.SCRIPTAMAT.2022.114603.
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- [21] ²⁰²² Hyunseok Kim, Sangho Lee, Jiho Shin, **Menglin Zhu**, Marx Akl, Kuangye Lu, Ne Myo Han, Yongmin Baek, Celesta S Chang, Jun Min Suh, Ki Seok Kim, Bo In Park, Yanming Zhang, Chanyeol Choi, Heechang Shin, He Yu, Yuan Meng, Seung Il Kim, Seungju Seo, Kyusang Lee, Hyun S Kum, Jae Hyun Lee, Jong Hyun Ahn, Sang Hoon Bae, Jinwoo Hwang, Yunfeng Shi, and Jeehwan Kim. Graphene nanopattern as a universal epitaxy platform for single-crystal membrane production and defect reduction. *Nat. Nanotechnol.*, volume 17, pages 1054–1059, DOI:10.1038/s41565-022-01200-6.
- [20] ²⁰²² Md Rezaul Karim, Benthara Hewage Dinushi Jayatunga, Kaitian Zhang, **Menglin Zhu**, Jinwoo Hwang, Kathleen Kash, and Hongping Zhao. Band Structure Engineering Based on $\text{InGaN}/\text{ZnGeN}_2$ Heterostructure Quantum Wells for Visible Light Emitters. *Cryst. Growth Des.*, volume 22, pages 131–139, DOI:10.1021/acs.cgd.1c00630.
- [19] ²⁰²² I Hsuan Kao, Ryan Muzzio, Hantao Zhang, **Menglin Zhu**, Jacob Gobbo, Sean Yuan, Daniel Weber, Rahul Rao, Jiahua Li, James H Edgar, Joshua E Goldberger, Jiaqiang Yan, David G Mandrus, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Deterministic switching of a perpendicularly polarized magnet using unconventional spin–orbit torques in WTe_2 . *Nature Materials* 2022 21:9, volume 21, pages 1029–1034, DOI:10.1038/s41563-022-01275-5.
- [18] ²⁰²¹ Tiancong Zhu, Alexander J Bishop, Tong Zhou, **Menglin Zhu**, Dante J O'Hara, Alexander A Baker, Shuyu Cheng, Robert C Walko, Jacob J Repicky, Tao Liu, and Others. Synthesis, Magnetic Properties, and Electronic Structure of Magnetic Topological Insulator MnBi_2Se_4 . *Nano Lett.*, DOI:10.1021/acs.nanolett.1c00141.
- [17] ²⁰²¹ Md Rezaul Karim, Brenton A Noesges, Benthara Hewage Dinushi Jayatunga, **Menglin Zhu**, Jinwoo Hwang, Walter R L Lambrecht, Leonard J Brillson, Kathleen Kash, and Hongping Zhao. Experimental determination of the valence band offsets of ZnGeN_2 and $(\text{ZnGe})_{0.94}\text{Ga}_{0.12}\text{N}_2$ with GaN. *J. Phys. D Appl. Phys.*, volume 54, page 245102, DOI:10.1088/1361-6463/abee45.

- [16] ²⁰²¹ Soohyun Im, Yunzhi Yuchi Wang, Pengyang Zhao, Geun Hee Yoo, Zhen Chen, Gabriel Calderon, Mehrdad Abbasi Gharacheh, **Menglin Zhu**, Olivia Licata, Baishakhi Mazumder, David A Muller, Eun Soo Park, Yunzhi Yuchi Wang, and Jinwoo Hwang. Medium-range ordering, structural heterogeneity, and their influence on properties of Zr-Cu-Co-Al metallic glasses. *Physical Review Materials*, volume 5, pages 1–11, DOI:10.1103/PhysRevMaterials.5.115604.
- [15] ²⁰²¹ Yang Cheng, Sisheng Yu, **Menglin Zhu**, Jinwoo Hwang, and Fengyuan Yang. Tunable topological Hall effects in noncollinear antiferromagnet Mn₃Sn/Pt bilayers. *APL Materials*, volume 9, page 51121, DOI:10.1063/5.0048733.
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- [11] ²⁰²⁰ Aidan J Lee, Adam S Ahmed, Brendan A McCullian, Side Guo, **Menglin Zhu**, Sisheng Yu, Patrick M Woodward, Jinwoo Hwang, P Chris Hammel, and Fengyuan Yang. Interfacial Rashba-Effect-Induced Anisotropy in Nonmagnetic-Material–Ferrimagnetic-Insulator Bilayers. *Phys. Rev. Lett.*, volume 124, page 257202, DOI:10.1103/PhysRevLett.124.257202.
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Teaching Experience

- Fall, 2023 **3.34 Imaging of Materials**, guest lecturer, Massachusetts Institute of Technology
- Fall, 2020 **MSE3151 Transport Phenomenon and Kinetics**, teaching assistant, Ohio State University
- Fall, 2019 **MSE3332 Undergraduate Lab II**, laboratory assistant and instructor, Ohio State University

Synergistic Activities

- 2023-present **Mentor of two graduate students** on project *Collaborative for Hierarchical Agile and Responsive Materials (CHARM) under cooperative agreement W911NF-19-2-011*
- 2023 **Co-organize Microscopy and Microanalysis conference P08 symposium** entitled *Atomic Scale Microscopy of Interfaces and Heterostructures with Correlated Phenomena*
- 2023 **Mentor of two REU Students** on project *NFO Thin Films Grown via an Off-Axis Sputtering Method; STEM Characterization of NdFeO₃/SrTiO₃ Thin Films*
- 2022 **Mentor of one REU Student** on project *Effects of Octahedral Tilting and Lattice Strain on LaFeO₃*

References

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